TUSHAR GAUTAM

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EDUCATION

Invertis University *Bareilly, Uttar Pradesh*
Bachelor of Technology in Computer Science Engineering *Aug 2022 - Jun 2026*

WORK EXPERIENCE

Software Engineer, Ethara AI *May 2026 - Present*
Gurgaon, Haryana

- Contribute to the **Long Horizon Terminal Benchmarking (LHTB) project**, evaluating AI agent performance on complex software engineering tasks through repository-level benchmarks, terminal execution and automated test validation.
- Work with Docker-based benchmarking pipelines to build repository-specific images, configure dependencies, apply code patches and execute isolated test environments across software repositories.
- Evaluate AI-generated code fixes against ground-truth patches using structured test execution workflows, analyzing baseline, test-patch and fix-patch results to measure task resolution and functional correctness.
- Analyze automated test outcomes and execution logs, transforming raw outputs into structured test results, including passed, failed, skipped and error states.
- Assess benchmark validity through test-state transition analysis, including fail-to-pass (f2p), skip-to-pass (s2p), none-to-pass (n2p) and pass-to-pass (p2p), identifying resolved issues, regressions and unsuccessful fixes.
- Support evaluation harnesses and reporting workflows for model-generated code patches, validating test outcomes, detecting regressions and assessing the reliability of AI-driven software engineering agents.

PROJECTS / OPEN-SOURCE

BrandBot AI Agent *Python, Fast API, Langchain, React.js, Tailwind CSS*

- Built an autonomous LinkedIn AI agent using LangChain, LangGraph, and Google Gemini that automates brand analysis, trend research and content creation.
- Designed a multi-agent workflow with 5+ specialized agents, integrated Tavily Search for real-time insights and reduced manual content ideation effort by ∼70%.

Snap UI *Next.js, React.js, Typescript, Tailwind CSS*

- Developed a powerful AI-driven UI builder that enables users to generate and customise modern web interfaces effortlessly.
- Implemented dynamic component rendering, real-time editing, and AI-assisted design suggestions using Next.js, Tailwind CSS and Convex backend.
- Automated workflows for auth, component generation, and previews, reducing manual setup effort by ∼60%.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, C.
GenAI / Agentic AI: Agentic AI, LLM Evaluation, AI Agent Benchmarking, Reinforcement Learning Trajectory Analysis, Model-Generated Code Evaluation, Evaluation Harnesses, Benchmarking Pipelines.
Tools / Platforms: Git/GitHub, Docker, Kubernetes, AWS, GitHub Actions, Linux, Terminal-Based Workflows, Automated Test Execution, Regression Detection, Test Result Analysis.
Backend / Databases: Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, MySQL, Redis, Firebase.
Libraries / Frameworks: React.js, Next.js, Mongoose, Tailwind CSS, Postman.